BTS-I(R&S)/II(S)-11.17-1592 Reg.No. [ ] **C**

B.Tech. Degree I Semester Regular and Supplementary/ II Semester Supplementary Examination November 2017

CE/EE/ME/SE GE 15-1103 A AND CS/EC/IT GE 15-1203 B
ENGINEERING MECHANICS
(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A
(Answer *ALL* questions)

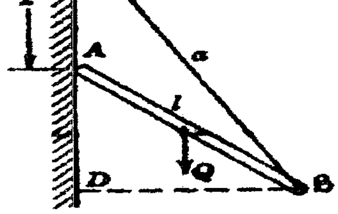
(10 × 2 = 20)

I. (a) What are the characteristics of a force? State the classification of force systems.
(b) Define resultant of a force system. State the law of parallelogram of forces.
(c) State Pappus's theorems of areas and volumes of revolution. Prove any one of them.
(d) Derive the second moment of area of a circular surface with respect to a tangent (to the circumference).
(e) Define the principle of virtual work. Explain its usefulness in solving problems of rigid body mechanics.
(f) Derive the impulse-momentum principle from the differential equation of motion (Newton's second law of motion). Consider a particle undergoing rectilinear translation.
(g) What is coefficient of restitution? Derive the basic equations for elastic, plastic and semi-elastic direct central impact.
(h) State D' Alembert's principle in curvilinear translation.
(i) Derive the principle of moment of momentum in curvilinear translation.
(j) Derive the time period of oscillations of a compound pendulum from fundamentals.

PART B

(4 × 10 = 40)

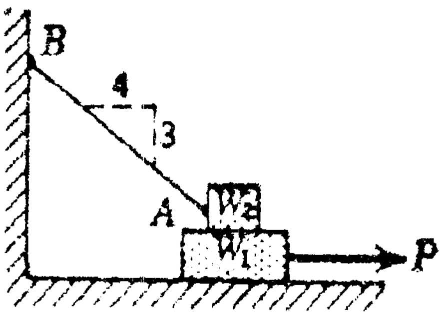
II. A prismatic bar of weight, 'Q' and length 'l' is supported at one end by a string of length 'a' and rests at A, vertically below C, against a perfectly smooth vertical wall (figure). Find the position of the bar as defined by the length, 'x' for which equilibrium will be possible. (10)



OR

(P.T.O.)

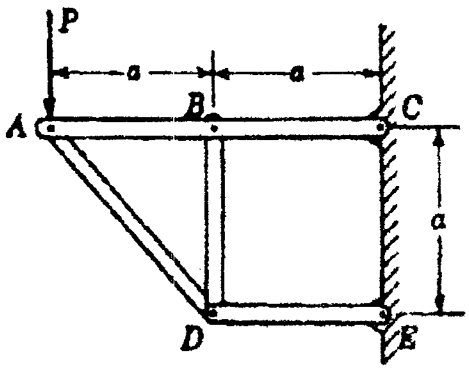
III. A block of weight $W_1 = 200\ N$ rests on a horizontal surface and supports on top of it another block of weight $W_2 = 50\ N$ (figure). The block W_2 is attached to a vertical wall by the inclined string AB. Find the horizontal force P applied to the lower block as shown, that will be necessary to cause slipping to impend. The coefficient of static friction for all contiguous surfaces is $\mu = 0.3$. (10)



IV. Locate the principal axes passing through the centroid of a 'T' section which has width of flange = $120\,mm$, depth of web = $150\,mm$ and thickness of flange and web = $12\ mm$, each. Also calculate the principal second moments of area. (10)

OR

V. Determine the axial force in the vertical member BD of the plane frame, supported and loaded as shown in the figure. (10)



VI. (a) A small block of weight W rests on an adjustable inclined plane. Friction is such that sliding of the block impends when the angle of inclination of the plane with the horizontal, $\alpha = 30°$. What acceleration will the block have when $\alpha = 45°$? Neglect any difference between static and kinetic friction. (5)

(b) Two men of masses $50\ kg$ and $75\ kg$ dive off the end of a boat of mass $250\ kg$ so that their relative velocity with respect to the boat is $4\ m/s$. If the boat is initially at rest, find its final velocity if (i) two men dive simultaneously (ii) the man of mass $75\ kg$ dives first followed by the other and (iii) the man of mass $50\ kg$ dives first followed by the other. (5)

OR

VII. A bullet of mass 0.01 kg moving with a velocity of 100 m/s, hits the bob of a simple pendulum, horizontally. Find the maximum angle through which the pendulum will swing when (i) the bullet gets embedded in the bob (ii) the bullet rebounds from the surface of the bob with a velocity of 20 m/s and (iii) the bullet escapes from the other end of the bob with a velocity of 20 m/s. The bob weighs 1 kg and the length of the string of the pendulum is 1 m. (10)

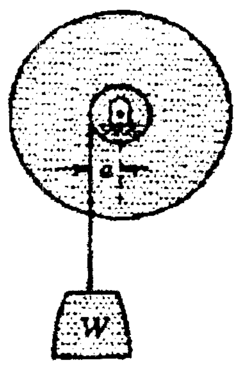
VIII. If coefficient of friction, $\mu = 0.2$ between the wheel and the road, find at what speed a vehicle can get round a curve of 50 m radius without side-slip (i) on a level road, (ii) on a road banked to a slope of 1 vertical to 8 horizontal and (iii) at what speed can the vehicle travel on the banked road without any lateral frictional support? (10)

OR

IX. The rotor and shaft in the figure weigh 120 N and the radius of gyration with respect to the axis of rotation is 25 cm. (10)

(i) Calculate the acceleration of the falling weight, $W = 25\ N$ if the shaft radius, $a = 12\ cm$.

(ii) Calculate the moment that must be applied to the shaft of the rotor to produce an upward acceleration of $\frac{g}{3}$ to the attached weight W.



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